

VLSI Internship- Task 1

Part A: Understanding VLSI & Its Applications

Introduction

VLSI stands for Very Large Scale Integration. It is the process of integrating millions of transistors onto a single silicon chip. VLSI technology is used in smartphones, computers, embedded systems, GPUs, and AI accelerators.

Examples of VLSI chips:

- i. Microprocessors
- ii. Mobile SoCs
- iii. Memory chips

Applications of VLSI:

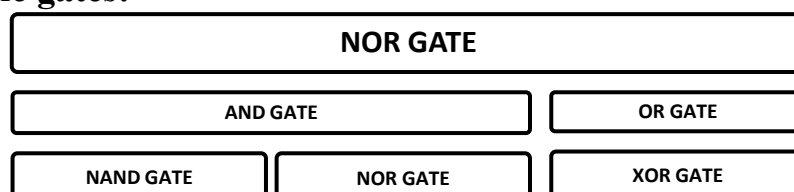
- Smartphones
- Computers & Laptops
- Automotive Electronics
- Medical devices
- AIML hardware
- IoT & Embedded Systems

Part B: VLSI Design Flow

1. System Specification
2. Architecture Design
3. RTL Design
4. Functional Verification
5. Synthesis
6. Physical Design
7. Timing and Power Analysis
8. Fabrication
9. Testing and Validation

Part C: Digital Logic Fundamentals

Basic logic gates:



LOGIC TRUTH TABLE:

AND GATE: $Y=AB$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR GATE: $Y=A+B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT GATE: $Y=\sim A$

A	Y
0	1
1	0

NAND GATE: $Y=\sim(AB)$

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR GATE: $Y=\sim(A+B)$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

XOR GATE: $Y=A\oplus B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

HALF ADDER: $SUM = A \oplus B$, $CARRY = AB$

A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

FULL ADDER: $SUM = A \oplus B \oplus C$, $CARRY = AB + BC + CA$

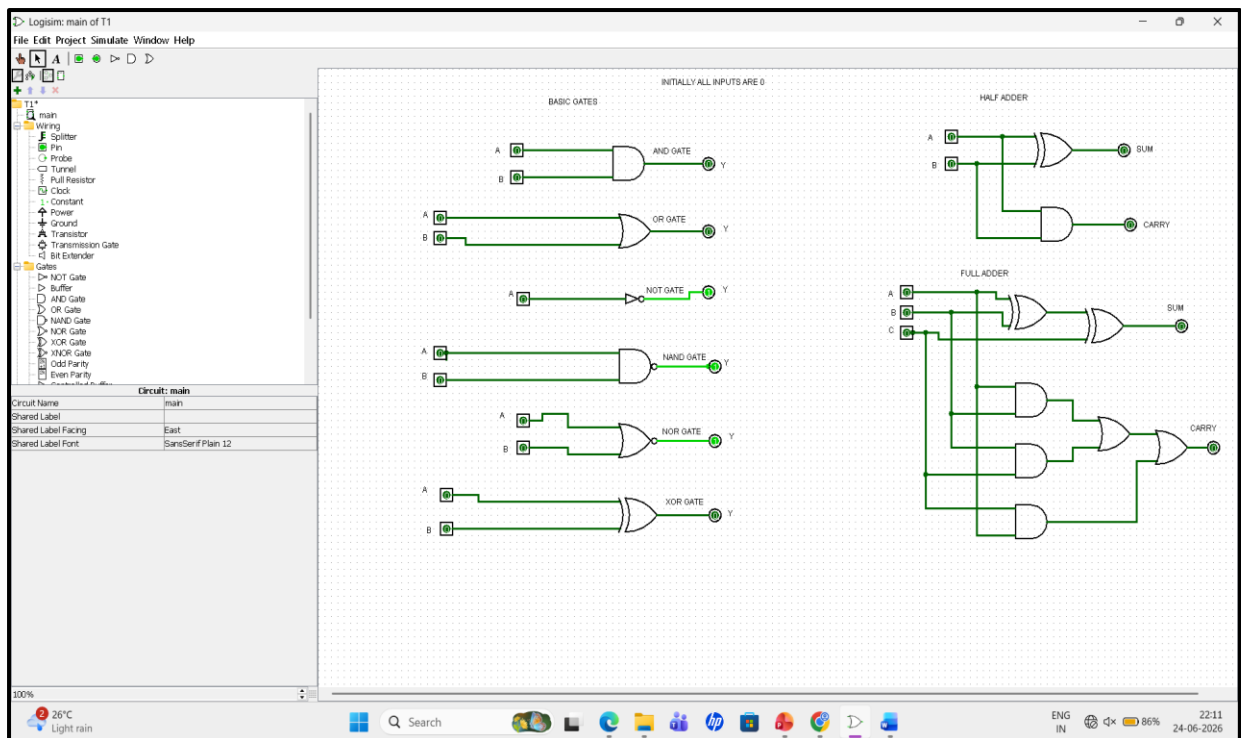
A	B	C	SUM	CARRY
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Part D: Practical Implementation (Simulation Task)

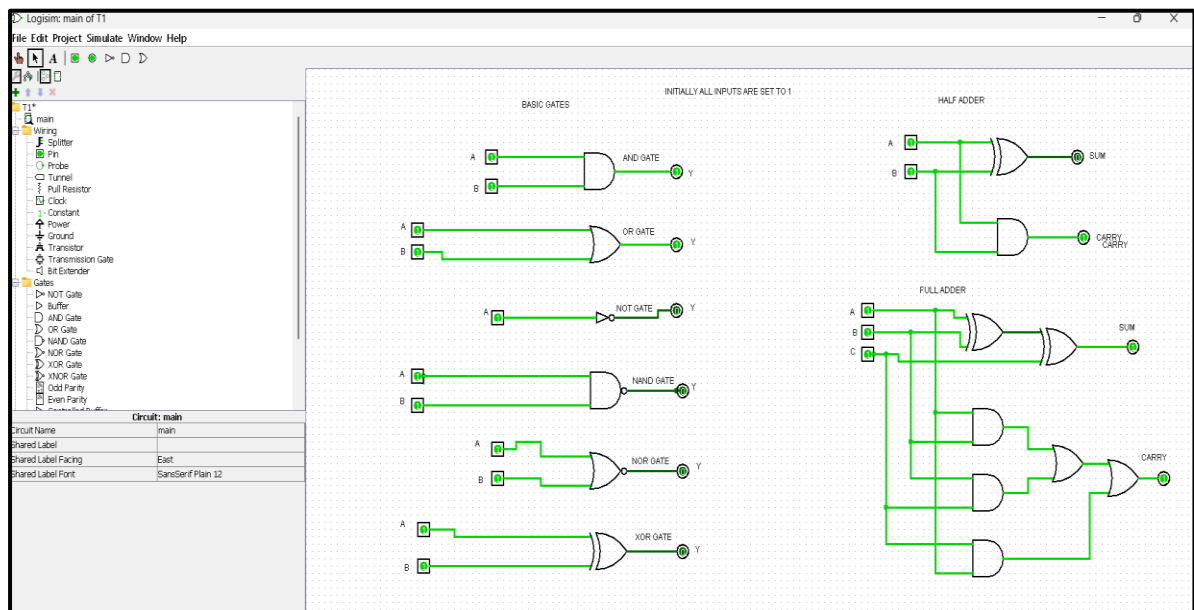
Software: Logisim

SCREENSHOTS

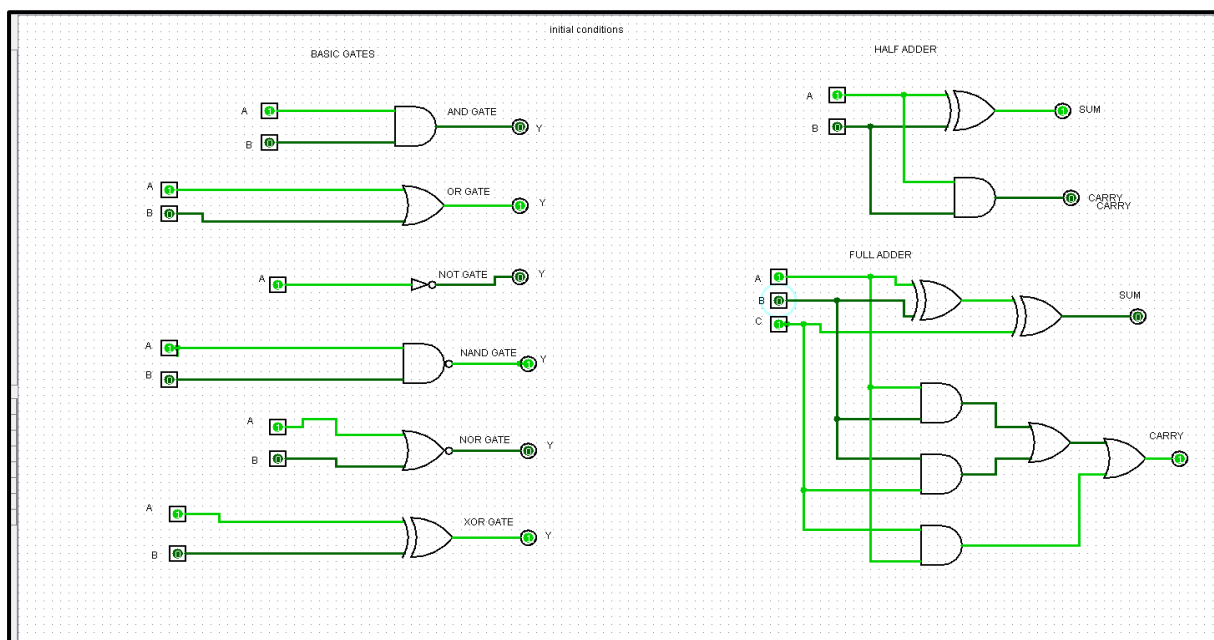
- INITIALLY ALL INPUTS ARE SET TO 0



- INITIALLY ALL INPUTS ARE SET TO 1



- INITIAL CONDITIONS
A=1 B=0 (For basic gates & half adder) & C=1 (for Full adder)



Learning Outcomes:

- ✓ Understood VLSI Fundamentals
- ✓ Learned VLSI design flow
- ✓ Implemented Logic gates & verified their truth tables accordingly
- ✓ Simulated Basic gates, Half adder & Full adder using Logisim & verified their outputs

Author

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